

# IEOR 4404: Simulation

## Lecture 6: Composition Method

# Composition Method

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- Generate random variables with “layered” structure, or “mixtures”
- Discrete mixtures:

$$P(X \leq x) = \sum_{j=1}^m P(Y_j \leq x)P(J = j)$$

or

$$P(X = x) = \sum_{j=1}^m P(Y_j = x)P(J = j)$$

- $X$  is a  $J$ -mixture of  $Y_1, \dots, Y_m$
- Law of total probability: Conditional on  $J = j$ , distribution of  $X$  follows that of  $Y_j$

# Composition Method

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To generate  $X$ ,

- Generate  $J$  (using ITM for example)
- Given  $J = j$ , generate  $Y_j$

The output follows the correct distribution

# Example

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$X$  is a discrete r.v. with pmf

$$P(X = x) = \begin{cases} 0.05, & x = 1, 2, \dots, 5 \\ 0.15, & x = 6, 7, \dots, 10 \end{cases}$$

Generate  $X$

# Example

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Composition method:

- Generate  $J$  where  $P(J = 1) = 0.25, P(J = 2) = 0.75$
- If  $J = 1$ , generate  $Y_1$  uniform over  $\{1, 2, \dots, 5\}$ ; if  $J = 2$ , generate  $Y_2$  uniform over  $\{6, 7, \dots, 10\}$

Pseudo code:

- Generate  $U \sim Unif(0,1)$
- Generate  $V \sim Unif(0,1)$
- If  $U \leq 0.25$ , set  $X = \lfloor 5V \rfloor + 1$ ; else set  $X = \lfloor 5V \rfloor + 6$

Is this the only way to use the composition method in this example?

# Example

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We can also let:

- $P(J = 1) = 0.5, P(J = 2) = 0.5$
- If  $J = 1$ , generate  $Y_1$  uniform over  $\{1, 2, \dots, 10\}$ ; if  $J = 2$ , generate  $Y_2$  uniform over  $\{6, 7, \dots, 10\}$

We can check that the distribution of the output is still correct

# Example

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Composition method:

- Generate  $J$  where  $P(J = 1) = 0.5, P(J = 2) = 0.5$
- If  $J = 1$ , generate  $Y_1$  uniform over  $\{1, 2, \dots, 10\}$ ; if  $J = 2$ , generate  $Y_2$  uniform over  $\{6, 7, \dots, 10\}$

Pseudo code:

- Generate  $U \sim Unif(0,1)$
- Generate  $V \sim Unif(0,1)$
- If  $U \leq 0.5$ , set  $X = \lfloor 10V \rfloor + 1$ ; else set  $X = \lfloor 5V \rfloor + 6$

# Composition Method

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- In the continuous case, we consider a mixture of a “continuum” set of random variables
- Continuous mixture:

$$P(X \leq x) = \int_0^{\infty} P(Y_t \leq x) f_T(t) dt$$

or density

$$f(x) = \int_0^{\infty} f_{Y_t}(x) f_T(t) dt$$

- $X$  is a  $T$ -mixture of  $\{Y_t: t \geq 0\}$



# Composition Method

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To generate  $X$ ,

- Generate  $T$
- Given  $T = t$ , generate  $Y_t$

The output follows the correct distribution

# Example

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Generate the two-sided exponential variable with density

$$f(x) = \frac{1}{2} e^{-|x|}, -\infty < x < \infty$$

# Cautions

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In the composition method, it is not necessary that the mixing variable and individual variables conditional on mixing are both discrete or continuous. Instead, it could be that one of them is discrete and another one continuous, or vice versa.

Sometimes the composition structure arises naturally from the problem, but sometimes not.

When the composition structure can be identified, the method can be faster than other more direct approaches, such as ITM and A-R.

## Example 2

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A production facility produces a Poisson number of products with mean  $t$ , in the time interval  $[0, t]$ . An inspector comes at a random time  $T \sim \text{Exp}(1)$ . Let  $X$  be the number of products that the inspector sees in  $[0, T]$ . Generate  $X$  using the composition method.

# Example 3

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Generate  $X$  with density

$$f(x) = \int_0^{\infty} 2e^{-\frac{x}{t}-t^2} dt$$

using the composition method